# Support - operational storage ER

Current amendment to ER item 1 - 2026-10-14 - Original ER pages/PDF archives retained

Queue/classification/responder fields live in support_tickets, not audit events.
SLA targets use a same-ticket/same-policy composite round FK (deferred).
UNIQUE(ticket_id, cycle); UNIQUE(sla_round_id, metric_type); source history is unique.
Policy snapshots keep recorded provenance. Unknown legacy values remain unknown.
Chat receipts contain metadata only. Chat retains ownership of message bodies.

### support_tickets

| | | |
|---|---|---|
| - | priority_score | INT NN |
| - | risk_report_count | INT NN |
| - | chat_responder_id | TEXT NULL |
| - | classification_reason_codes | TEXT[] NULL |
| - | classification_policy_version | TEXT NULL |
| - | risk_lookup_available | BOOL NULL |
| PK | ticket_id | TEXT NN |
| UQ | ticket_number | TEXT NN |
| - | requester_id | TEXT NN |
| - | subject | TEXT NN |
| - | description | TEXT NN |
| FK | ticket_category_id | SMALLINT NN |
| FK | ticket_priority_id | SMALLINT NN |
| FK | current_status_id | SMALLINT NN |
| - | order_id | TEXT NULL |
| - | target_user_id | TEXT NULL |
| - | lock_version | INT NN |
| - | created_at | TIMESTAMPTZ(3) NN |
| - | updated_at | TIMESTAMPTZ(3) NN |

### ticket_sla_rounds

| | | |
|---|---|---|
| PK | sla_round_id | TEXT NN |
| FK | ticket_id | TEXT NN |
| - | cycle | INT NN |
| - | started_at | TIMESTAMPTZ(3) NN |
| FK/UQ | source_history_id | TEXT NULL |
| FK | sla_policy_id | TEXT NN |
| - | policy_effective_from | TIMESTAMPTZ(3) NULL |
| - | policy_effective_to | TIMESTAMPTZ(3) NULL |
| - | first_response_minutes | INT NULL |
| - | resolution_minutes | INT NULL |
| - | snapshot_source | TEXT NN |

### ticket_sla_targets

| | | |
|---|---|---|
| FK | sla_round_id | TEXT NN |
| PK | sla_target_id | TEXT NN |
| FK | ticket_id | TEXT NN |
| FK | sla_policy_id | TEXT NN |
| - | metric_type | VARCHAR(30) NN |
| - | due_at | TIMESTAMPTZ(3) NN |
| - | achieved_at | TIMESTAMPTZ(3) NULL |
| - | breached_at | TIMESTAMPTZ(3) NULL |
| - | created_at | TIMESTAMPTZ(3) NN |
| - | updated_at | TIMESTAMPTZ(3) NN |

### ticket_chat_message_receipts

| | | |
|---|---|---|
| PK | chat_message_id | TEXT NN |
| FK | ticket_id | TEXT NN |
| - | conversation_id | TEXT NN |
| - | author_id | TEXT NULL |
| - | author_role | TEXT NN |
| - | is_internal | BOOL NN |
| - | message_created_at | TIMESTAMPTZ(3) NN |
| - | received_at | TIMESTAMPTZ(3) NULL |

### sla_policies

| | | |
|---|---|---|
| REF | sla_policy_id | TEXT NN |

### ticket_status_history

| | | |
|---|---|---|
| REF | status_history_id | TEXT NN |
| REF | ticket_id | TEXT NN |

## Relationship keys (physical FK unless marked logical)

**R1: ticket_sla_rounds -> support_tickets [parent:child 1 : 0..N]**

```
ticket_id -> ticket_id
```

**R2: ticket_sla_targets -> ticket_sla_rounds [parent:child 1 : 0..N]**

```
sla_round_id,ticket_id,sla_policy_id -> sla_round_id,ticket_id,sla_policy_id
```

**R3: ticket_sla_rounds -> sla_policies [parent:child 1 : 0..N]**

```
sla_policy_id -> sla_policy_id
```

**R4: ticket_sla_rounds -> ticket_status_history [parent:child 1 : 0..1 / optional]**

```
source_history_id,ticket_id -> status_history_id,ticket_id
```

**R5: ticket_chat_message_receipts -> support_tickets [parent:child 1 : 0..N]**

```
ticket_id -> ticket_id
```

PK primary key | FK foreign key | UQ unique | NN not null | IDs to other services are logical references